SEMESTER 3 · SECTIONS B & C

ANSWERING TECHNIQUE

Know it AND present it. Most lost marks come from presentation, not from the physics.

1 HOW MARKS ARE GIVEN

1. **Specific formula** (signs chosen)
2. **Substitution** with values
3. **Answer + unit**, correct s.f.

EXAMPLE · DOPPLER

× $f' = \dfrac{v \pm u_o}{v \mp u_s} f$ → no mark

✓ $f' = \dfrac{v + u_o}{v + u_s} f = \dfrac{340 + 250}{340 + 200}(16\,000)$

$= 17.5$ kHz

EXAMPLE · INTENSITY LEVEL

$\beta = 10 \log_{10} \dfrac{I}{I_0}$

$= 10 \log_{10} \dfrac{2.0 \times 10^{-6}}{1.0 \times 10^{-12}} = 63$ dB

2 SIGNIFICANT FIGURES & Π

- Use the **least s.f. in the data** (STPM accepts up to 4 s.f.).
- No answer **in terms of π** unless asked.

EXAMPLE · WAVE EQUATION

λ = 10 cm, f = 400 Hz, A = 2.0 cm

× $y = 0.020 \sin(20\pi x - 800\pi t)$

✓ $y = 0.020 \sin(63x - 2.5 \times 10^3 t)$ m

3 ERROR CARRIED FORWARD

A wrong answer from (a), used correctly in (b), still earns the marks in (b).

EXAMPLE

(a) λ = 0.50 m × (should be 0.40 m)

(b) $v = f\lambda = 850 \times 0.50 = 425$ m s^{-1} ✓ ECF

Never leave (b) and (c) blank because (a) went wrong.

4 STATE & DEFINE

Write a **full sentence** with the key words. A formula alone scores 0.

INTENSITY

× $I = P/A$

✓ The **rate of energy transferred per unit cross-sectional area perpendicular** to the direction of wave propagation.

BEATS

× $f = f_1 - f_2$

✓ **Periodic variation in intensity** caused by the superposition of two waves of **slightly different frequencies**.

5 EXPLAIN IN SEQUENCE

Ask after each point: **"If this happens, what follows next?"** Right points in the wrong order may score 0.

PROPAGATION OF SOUND

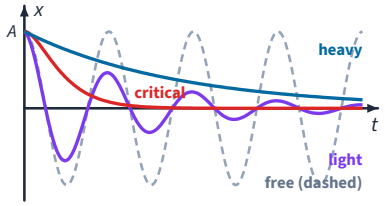
1. Air particles move **to and fro**.
2. **Compressions** (high pressure) and **rarefactions** (low pressure) form.
3. Particles vibrate **parallel** to the propagation: a longitudinal wave travels.

STANDING WAVE IN A PIPE

1. Air blown in: air particles vibrate.
2. Sound wave **reflected** at the end.
3. Incident and reflected waves **superpose**.
4. **Standing wave** with nodes and antinodes.

6 SKETCH GRAPHS

Label **both axes** with the quantity (units optional). Unlabelled = 0.



Comparing damping: **all curves start from the amplitude** (released from A in an experiment).

EXAMPLE · SOUND WAVE

On a displacement–distance graph, midway from **+max to −max** displacement = **compression** (max pressure); from −max to +max = **rarefaction** (min pressure).

7 NO MULTIPLE ANSWERS

Two alternative formulae or answers = **0 marks**. Decide first.

EXAMPLE · BEATS

P: 256 Hz, 4 beats s^{-1}. Loading Q **lowers** the beat frequency.

× $f_2 = 256 + 4$ or $256 - 4$

✓ Loading lowers f_2 and f_b drops, so $f_b = f_2 - f_1$:

$4 = f_2 - 256$, $f_2 = 260$ Hz

Changed your mind? Cross out the wrong working so only **one** answer remains.

8 CALCULATE SAFELY

Ionisation energy: start from $E_\infty - E_1$

$$E = 0 - (-13.6) = 13.6 \text{ eV}$$

Bohr energy: key in steps (m ÷ 8 ÷ ε_0^2, × e^4, ÷ h^2, ÷ n^2) or the calculator shows 0.

Use **1 u = 931.5 MeV** only if given; else $E = \Delta mc^2$ in kg.

9 READ, THEN CHOOSE

Thin film in air, reflected light **appears red** → blue is cancelled (destructive):

$$t_{min} = \frac{\lambda_b}{2n} = \frac{465}{2(1.33)} = 175 \text{ nm}$$

Not always λ/4n. **Malus:** first polariser gives $I_0/2$, then $I = I_1 \cos^2\theta$.

10 COMPARE QUESTIONS

	Beats	Interference
Waves	slightly different f	coherent
Observer	stays still	must move

	Progressive	Standing
Profile	moves	does not move
Energy	transferred	not transferred

EXAM TIPS

- ✓ No repeat paper for Sem 3: prepare fully.
- ✓ Analyse every diagram **before** the exam.
- ✓ Sem 1 & 2 skills are fair game (cosine rule: $I' = 3.7I_0$).
- ✓ Change in mass = final − initial (keep the sign).